

Algebra 1
Unit 6
Lesson 4 Homework

Name Answer Key
Date _____
Period _____

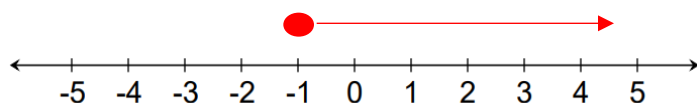
Consider the inequality $-2x + 5 \leq 7$.

1. Solve the inequality. $-2x + 5 \leq 7$
 $-5 \quad -5$

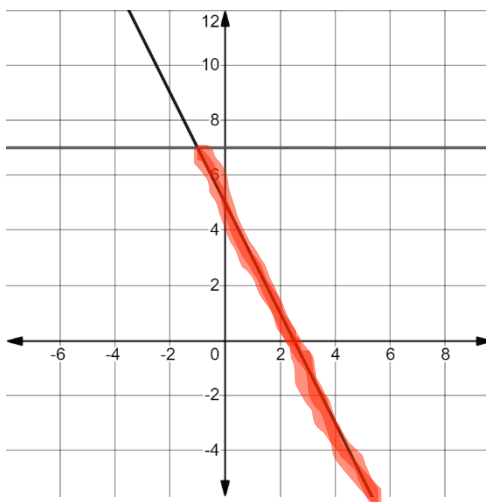
$$\frac{-2x}{-2} \leq \frac{2}{-2}$$

$$x \geq -1$$

2. Graph the solution set.



3. The graphs of $y = -2x + 5$ and $y = 7$ are shown on the coordinate plane.

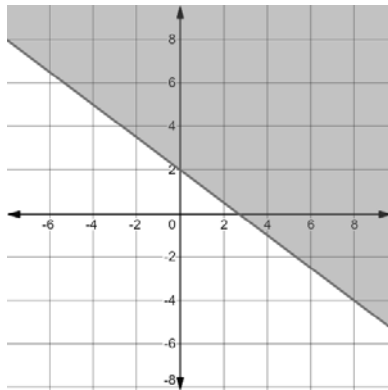


Highlight the part of the graph of $y = -2x + 5$, where the y values are less than or equal to 7.

4. Describe the relationship between the solution to the inequality $-2x + 5 \leq 7$ and the values of x for which the graph of $y = -2x + 5$ is less than or equal to 7.

When $x \geq -1$, the solution to $-2x + 5 \leq 7$, the y -values of $y = -2x + 5$ are ≤ 7 .

The graph of the two-variable inequality $y \geq -\frac{3}{4}x + 2$ is shown.



5. The slanted solid line is called the **boundary line** of the inequality. Write the equation of the boundary line in slope-intercept form.

$$y = -\frac{3}{4}x + 2$$

6. How does the equation of the boundary line relate to the inequality?

They have the same slope and y-intercept & are the same side from the $=$ & $\geq$.

7. Describe what part(s) of the graph represent the solution set of the inequality $y \geq -\frac{3}{4}x + 2$.

The shaded parts contains all values that represent solutions.

8. Which part of the graph of the solution set of $y \geq -\frac{3}{4}x + 2$ corresponds to each mathematical statement?

Mathematical Statement	Graphical Representation
$y = -3/4x + 2$	☒ boundary line ☐ shaded region
$y > -3/4x + 2$	☐ boundary line ☒ shaded region

Complete the statements.

9. If the points on the line are included in the solution set, the boundary line

is ☒ dotted.
☒ solid.

10. If the points on the line are not included in the solution set, the boundary

line is ☒ dotted.
☐ solid.